SC Algebra 1

Sub Day: Slope and Graphing Lines

Name: _____ Period: ___ Partner: _____

Work on your own or with your partner. Every part has a check: a code letter, a gold ★ your line must hit, or a check sum. If a check fails, fix it before you move on.

Quick review

Slope (rise over run): start at the FIRST point. Rise: how far from the first y to the second y? Run: how far from the first x to the second x? Up/right +, down/left −. Simplify.

Slope-intercept form y = mx + b: m = slope, b = y-intercept. Graph: PLACE (0, b), COUNT rise then run right, DRAW the line with arrows.

Special slopes: rise 0 → slope 0 (horizontal). Run 0 → undefined (vertical).

Part A — Slope between two points: crack the code

1. (1, 2) and (4, 8)	**2.** (−3, 5) and (1, −3)	**3.** (2, −1) and (6, 1)
4. (−4, −2) and (2, −6)	**5.** (5, 3) and (5, −4)	**6.** (−2, 7) and (6, 7)
7. (0, −5) and (3, 4)	**8.** (−6, 1) and (−1, −2)	**9.** (1, 1) and (3, 7)

What did the y-intercept say to the line?

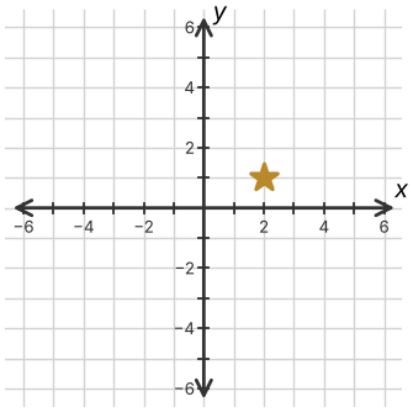
#1	#2	#3	#4	#5	#6	#7	#8	#9

Answer	Letter	Answer	Letter	Answer	Letter	Answer	Letter
2/3	U	0	H	2	S	−2	T
3/2	O	undefined	T	3	E	1/2	A
−2/3	R	−3/2	G	−3	L	−3/5	R
5/3	P	−1/2	N	1/3	D	−5/3	Y

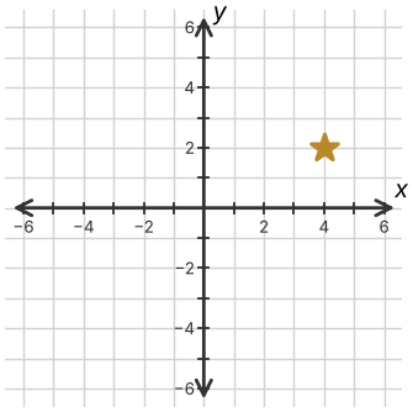
Part B — Graph each line

Place b, count the slope twice, draw the line with arrows. **✔ Check: your line must go right through the gold ★.**

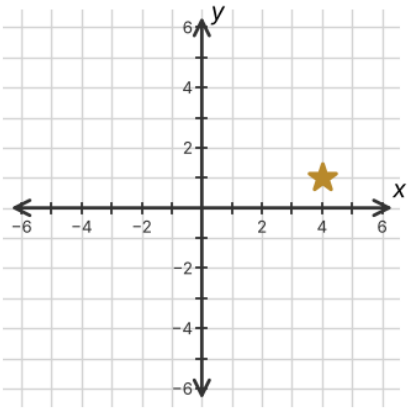
10. y = 2x − 3



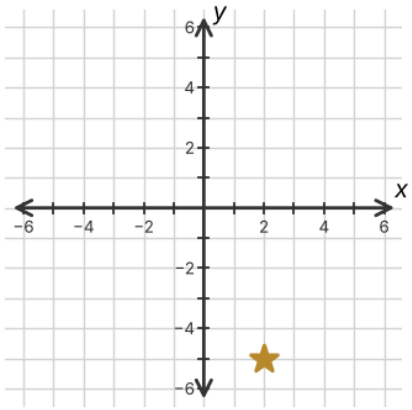
11. y = −(1/2)x + 4



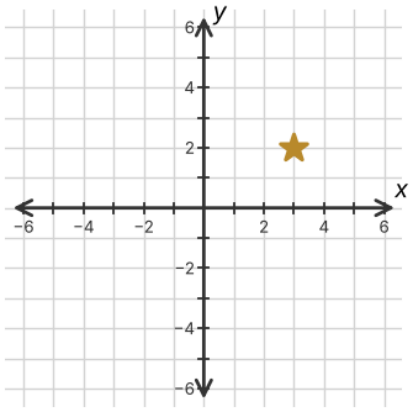
12. y = (3/4)x − 2



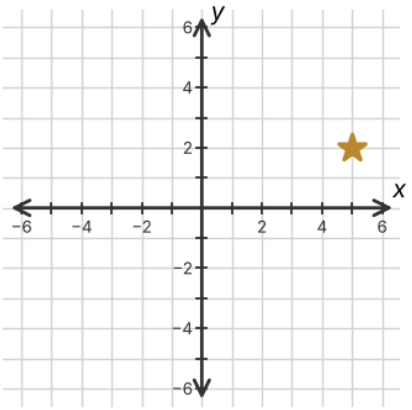
13. y = −3x + 1



14. y = 5 − x

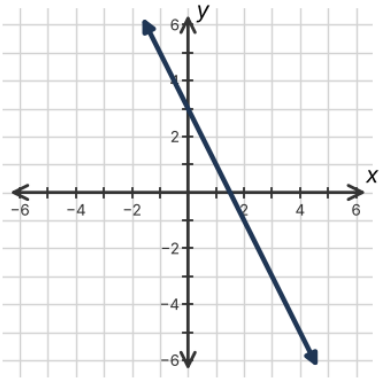


15. y = (2/5)x

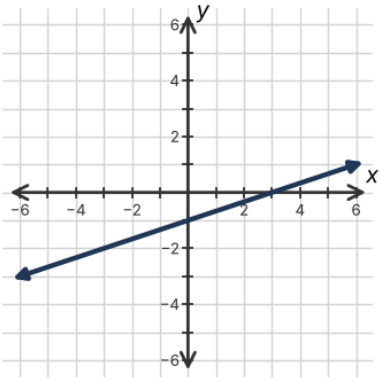


Part C — Write the equation of each line

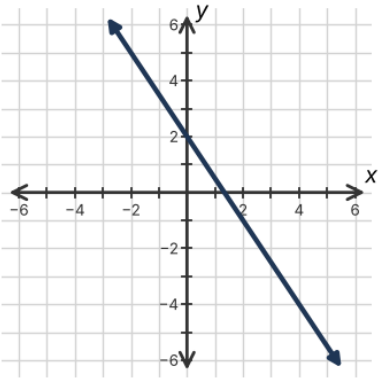
16. Write y = mx + b for the line.



17. Write y = mx + b for the line.



18. Write y = mx + b for the line.



✓ **Check yourself:** Your three b-values add up to 4. One slope is a negative whole number, one is a positive fraction, one is a negative fraction.

Part D — Slope in real life

19. A phone plan costs $20 with no data. With 5 GB of data it costs $45. Find the slope and say what it means. Then write the equation y = mx + b.

✓ **Check yourself:** Use your equation for 10 GB. It should cost $70.

20. A candle burns down steadily. After 0 hours it is 12 inches tall, after 2 hours 9 inches, after 4 hours 6 inches. Find the slope and say what it means. Write the equation.

✓ **Check yourself:** Set y = 0 in your equation. The candle should burn out at exactly 8 hours.

★ Challenge: Spiral review (Modules 1 – 3)

Everything so far. All answers are in the scrambled bank at the bottom (with a few traps). Cross each one off.

S1. Solve: $3(2x - 5) + 4 = 2x + 9$

S2. Solve for y, then name the slope and y-intercept: $4x - 2y = 10$

S3. Solve: $|2x - 3| = 9$

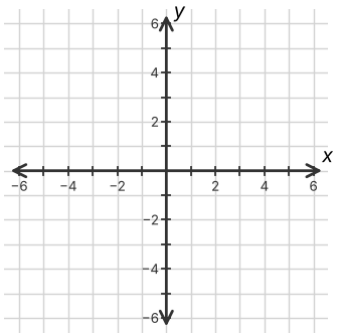
S4. Solve: $(x + 2)/6 = 5/3$

S5. Solve: $(3/4)x - 5 = 4$

S6. State the domain and range. Is it a function? {(−2, 4), (0, 1), (3, 4), (5, −2)}

S7. f(x) = −3x + 7. Find f(−2) − f(4).

S8. Find the x- and y-intercepts of 3x − 4y = 12, then use them to graph the line.



S9. ★ Boss level: Write the equation of the line through (2, 5) and (6, −3).

Scrambled answer bank (includes 5 traps)

x = 6 or x = −3 x = 8 y = −2x + 1 −6 x = 4 y = 2x − 5; m = 2, b = −5 18 (4, 0) and (0, −3) x = 5 y = −2x + 9 D {−2, 0, 3, 5}; R {−2, 1, 4}; yes (−3, 0) and (0, 4) x = 12 x = 6 only